

Mechatronics class notes

1. Electrical actuation systems: Mechanical switches (relays),
2. Solid-state switches,
3. Solenoids,
4. DC motors,
5. PWM and speed control,
6. AC motors,
7. Stepper motors,
8. DC and AC servo motors,
9. Tacho generators,
10. Synchros

Notes

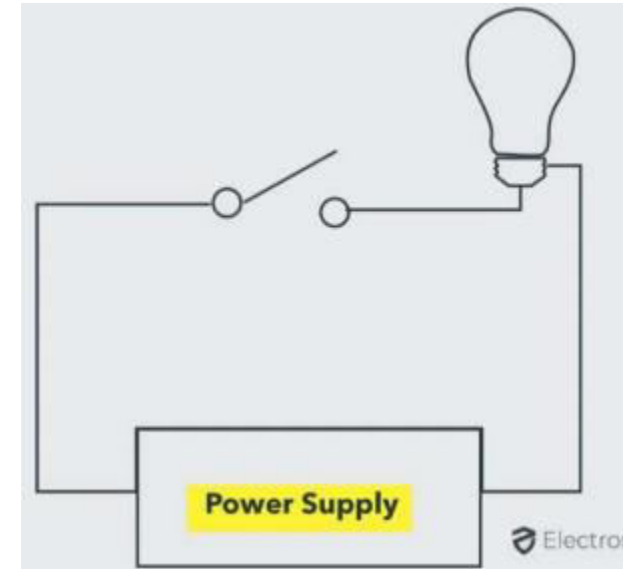
1. Electrical actuation systems: Mechanical switches (relays),
2. Solid-state switches,

Mechanical switches

- Mechanical Switches are **physical switches, which must be activated physically, by moving, pressing, releasing, or touching its contacts.**
- Mechanical switches classified based on
 - method of actuation (manual, limit and process switches),
 - number of contacts (single contact and multi contact switches),
 - Number of poles and throws {SPST,DPDT,SP OT, etc.},
 - Operation and construction (push button, toggle, rotary, Joystick, etc.),
 - based on state (momentary and locked switches),etc.

Single Pole Single Throw Switch (SPST)

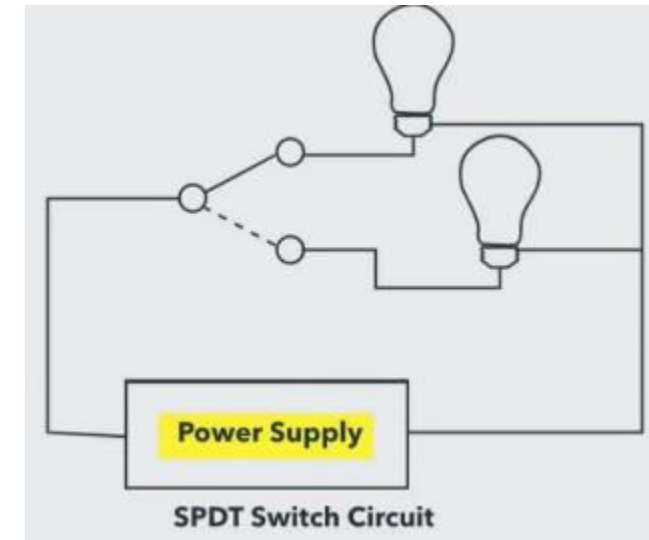
- This is the basic ON and OFF switch consisting of one input contact and one output contact.
- It switches a single circuit and it can either make (ON) or break (OFF) the load.
- The contacts of SPST can be either normally open or normally closed configurations .



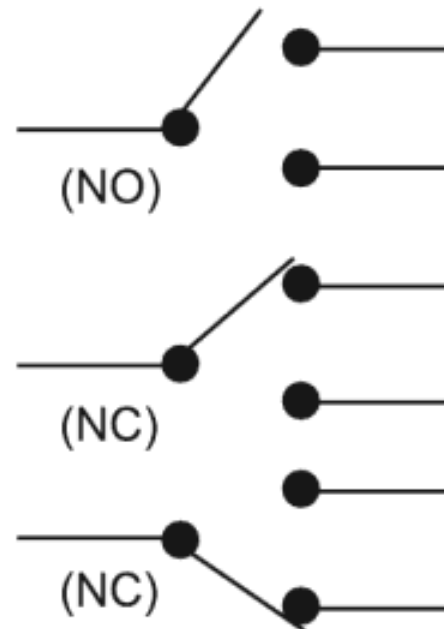
Type of Switches	Configuration	Contact Detail
1. Single Pole-Single Throw (SPST) contact	(NO) (NC)	SPST arrangement has one normally open (NO) and one normally closed (NC) contact.

Single Pole Double Throw Switch (SPDT)

- This switch has three terminals: one is input contact and remaining two are output contacts.
- This means it consists of two ON positions and one OFF position.
- In most of the circuits, these switches are used as changeover to connect the input between two choices of outputs.
- The contact which is connected to the input by default is referred to as a normally closed contact and the contact which will be connected during ON operation is a normally open contact.



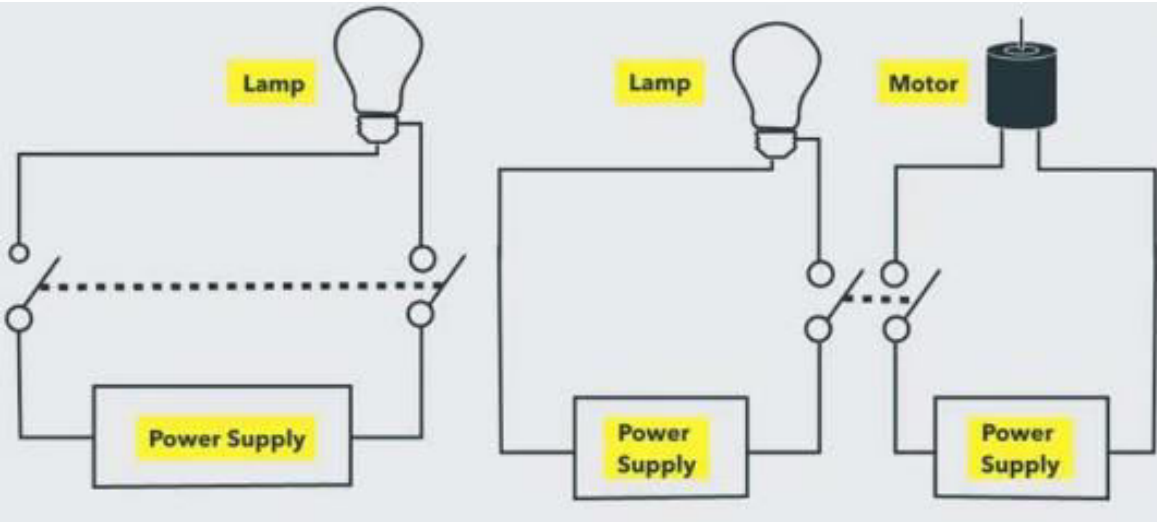
2. Single Pole-Double Throw (SPDT) contact



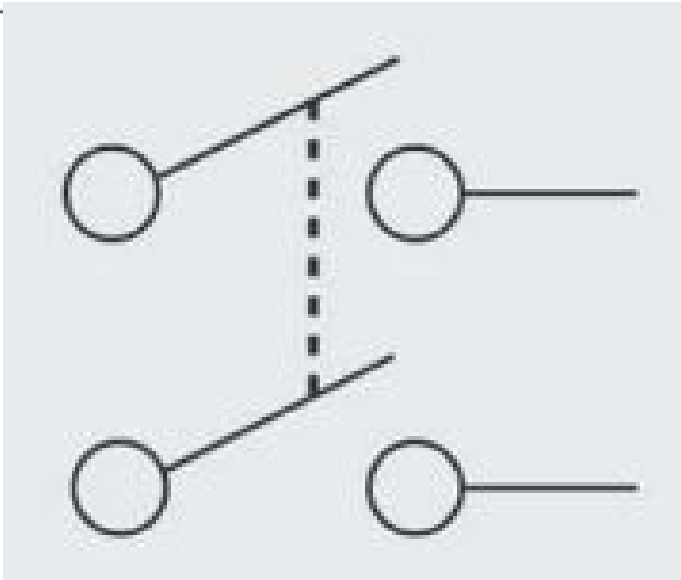
SPDT contact has one normally open (NO) and two normally closed (NC) arrangements possible.

Double Pole Single Throw Switch (DPST)

- This switch consists of four terminals: two input contacts and two output contacts.
- It behaves like a two separate SPST configurations, operating at the same time.
- It has only one ON position, but it can actuate the two contacts simultaneously, such that each input contact will be connected to its corresponding output contact.
- In OFF position both switches are at open state.
- This type of switches is used for controlling two different circuits at a time.
- Also, the contacts of this switch may be either normally open or normally closed configurations.



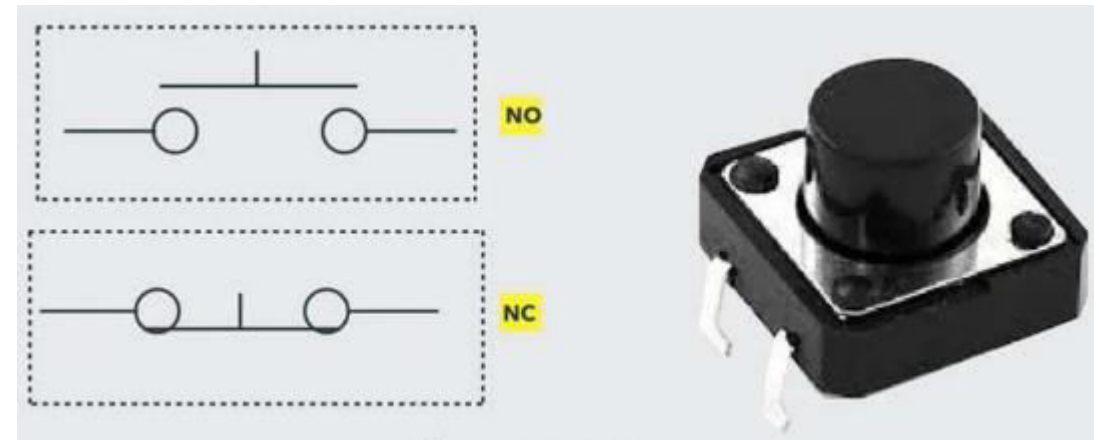
3. Double Pole-Double Throw (DPDT) contact



In this arrangement there are two normally open (NO) and two normally closed (NC) possibilities.

Push Button Switch

- It is a momentary contact switch that makes or breaks connection as long as pressure is applied (or when the button is pushed).
- Generally, this pressure is supplied by a button pressed by someone's finger.
- This button returns its normal position, once the pressure is removed.
- The internal spring mechanism operates these two states (pressed and released) of a push button.
- It consists of stationary and movable contacts, of which stationary contacts are connected in series with the circuit to be switched while movable contacts are attached with a push button.
- Push buttons are majorly classified into normally open, normally closed and double acting push buttons as shown in the above figure.
- Double acting push buttons are generally used for controlling two electrical circuits.



Toggle Switch

- A toggle switch is manually actuated (or pushed up or down) by a mechanical handle, lever or rocking mechanism. These are commonly used as light control switches.
- Most of these switches come with two or more lever positions which are in the versions of SPOT, SPST, DPST and DPDT switch. These are used for switching high currents (as high as 10 A) and can also be used for switching small currents
- These are available in different ratings, sizes and styles and are used for different type of applications. The ON condition can be any of their level positions, however, by convention the downward is the closed or ON position.

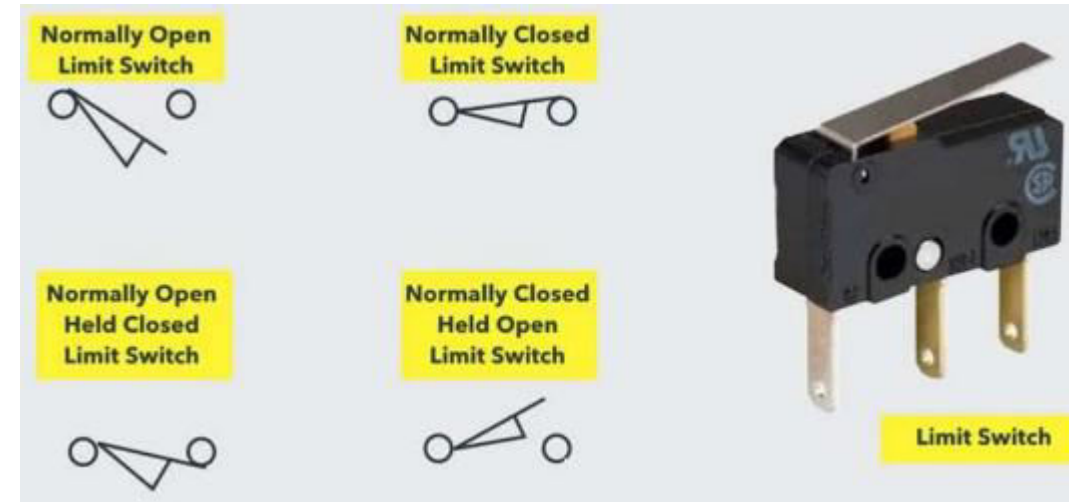


Limit Switch

- The control schemes of a limit switch are shown in above figure in which four varieties of limit switches are presented.
- Some switches are operated by the presence of an object or by the absence of objects or by the motion of machine instead of human hand operation. These switches are called as limit switches.

How it works

- These switches consist of a bumper type of arm actuated by an object. When this bumper arm is actuated, it causes the switch contacts to change position.

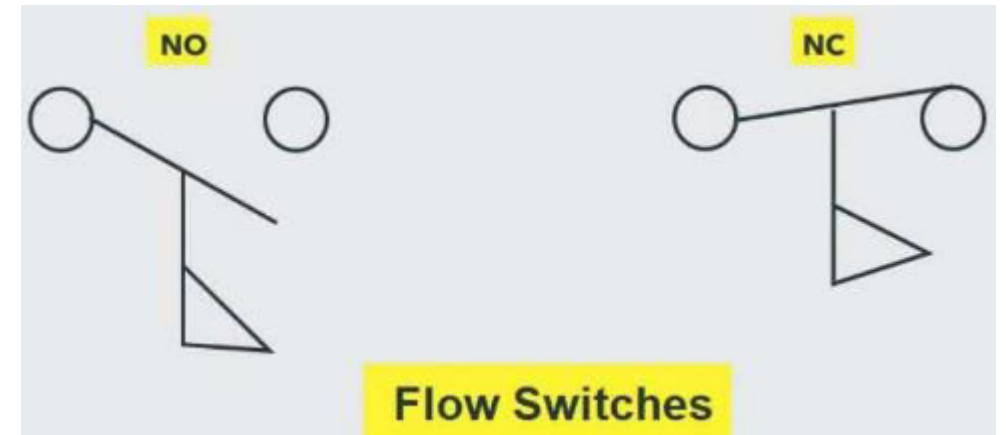


Flow Switches

- These are mainly used to detect the movement of liquid or air flow through a pipe or duct. The air flow switch (or a micro switch) is constructed by a snap action.

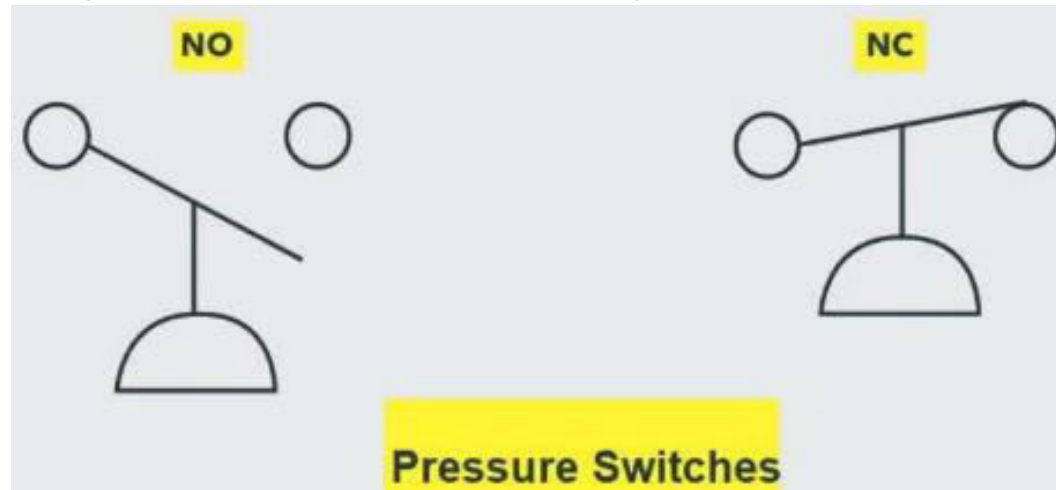
How it works

- The micro switch is attached to a metal arm .To this metal arm, a thin plastic or metal piece is connected. When a large amount of air passes through the metal or plastic piece, it causes the movement of metal arm and thus operates the contacts of the switch.
- Liquid flow switches are designed with a paddle that inserted across the flow of liquid in a pipe. When liquid flows through the pipe, **force exerted against the paddle changes the position of the contacts.**
- The above figure shows the switch symbol used for both air flow and liquid flow. The flag symbol on the switch indicates the paddle which senses the flow or movement of liquid.
- These switches again normally open or normally closed type configurations.



Pressure Switches

- ❑ These switches are commonly used in industrial applications in order to sense the pressure of hydraulic systems and pneumatic devices.
- ❑ Depends on the range of pressure to be measured, these pressure switches are classified into diaphragm operated pressure switch, metal bellow type pressure switch and piston type pressure switch.
- ❑ In all these types, pressure detection element operates a set of contacts (which can be either double pole or single pole contacts).
- ❑ This switch symbol consist a half-circle connected to a line in which flat part indicates a diaphragm. These switches may be either normally open or normally closed type configurations.

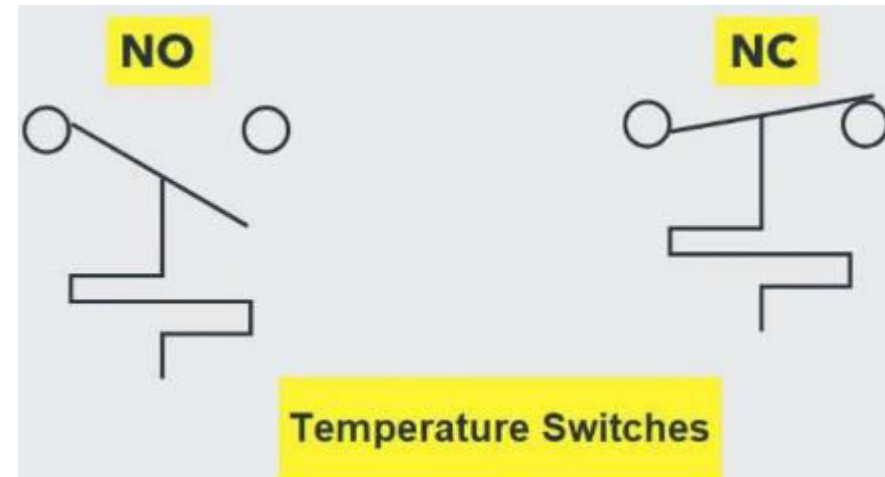


Temperature Switches

- The most common heat sensing element is the bimetallic strip that operates on the principle of thermal expansion.

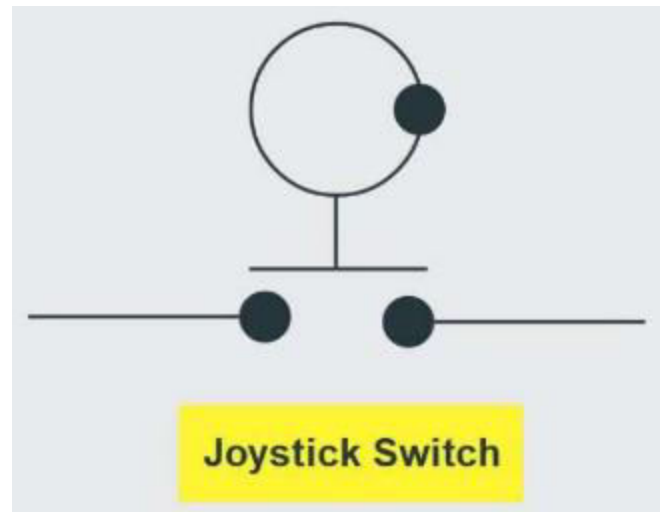
How it works

- The bimetallic strips are made with two dissimilar metals {that are having different thermal expansion rates) and are bonded with each other. The switch contacts are operated when the temperature causes the strip to bend or wrap.
- Another method of operating the temperature switch is to use mercury glass tube. When the bulb is heated, mercury in the tube will expand and then generates pressure to operate the contacts.

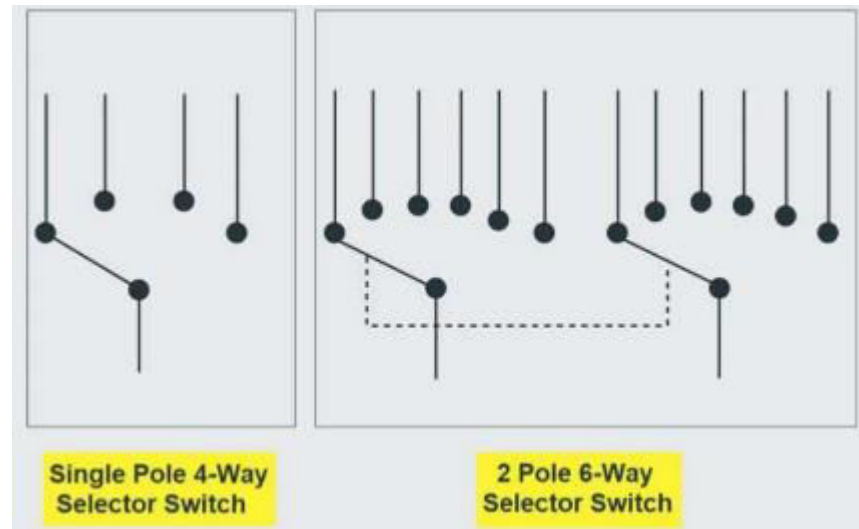


Joystick Switch

- Joystick switches are manually actuated control devices used mainly in portable control equipment.
- It consists of a lever which moves freely in more than one axis of motion.
- Depending on the movement of the lever pushed, one or more switch contacts are actuated.
- These are ideally suited for lowering, raising and triggering movements to the left and right.
- These are used for building machinery, cable controls and cranes. The symbol for the joystick is shown below.

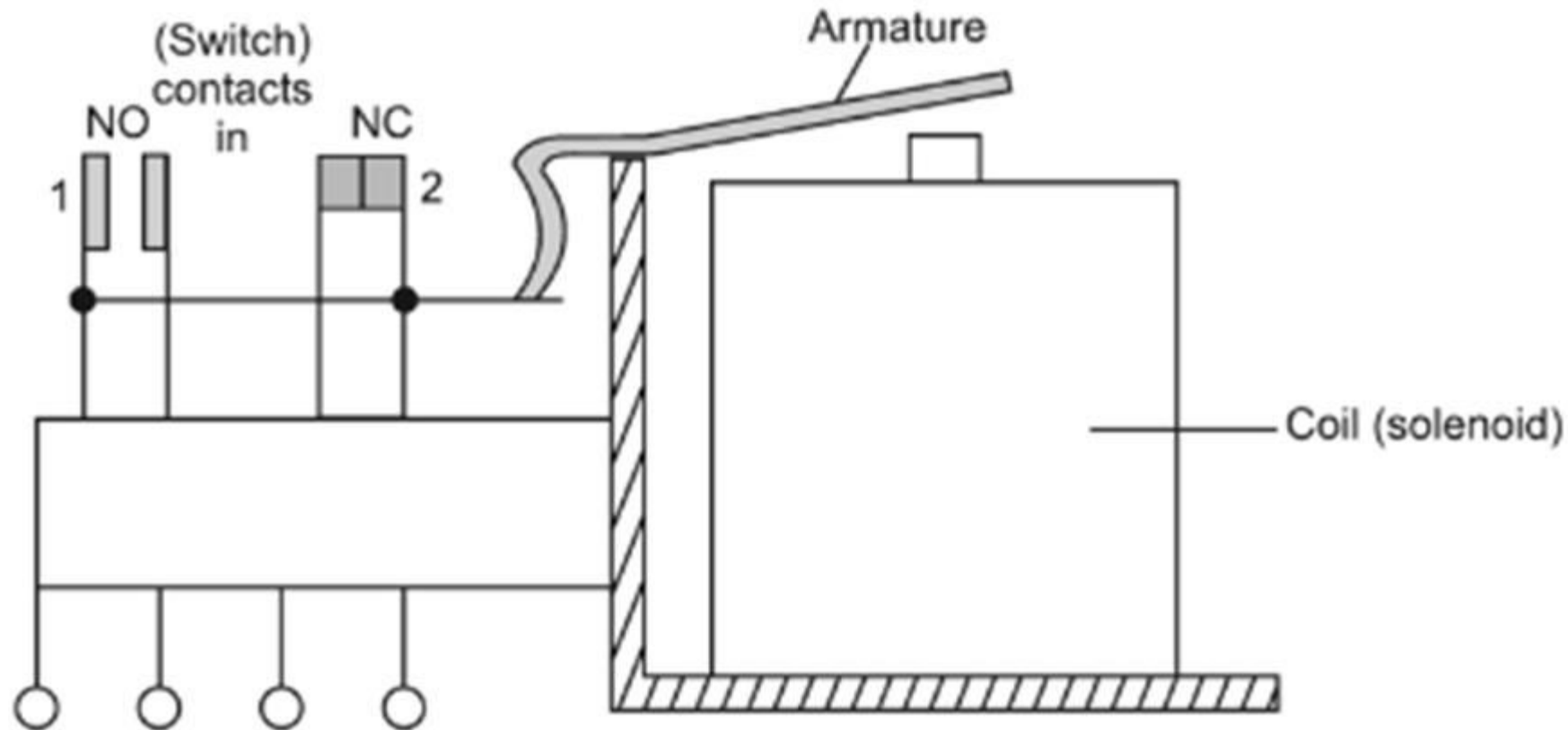


- **Rotary Switches**
- These are used for connecting one line to one of many lines.
- Examples of these switches are range selectors in electrical metering equipment, channel selectors in communication devices and band selectors in multi band radios.
- It consists of one or more moving contacts (knob) and more than one stationary contact.
- These switches are come with different arrangement of contacts such as single pole 12-way:, 3-pole 4-way,. 2-pole 6-way and 4-pole 3-way.



RELAYS

- Relays are basically switches that have electrical operation. By a change of the current in one leg of the circuit the other leg of the circuit is switched on or off the current. The schematic of a relay is in Figure



Relays in construction consist of a solenoid with a coil and armature. The armature is connected to switches with contact pairs. When there is no current in the coil of the solenoid the armature is not in touch with the plunger. The Switch 1 is in the normally open (NO) state and switch 2 is in the normally closed (NC) state. The flow of current through the coil magnetizes the coil and the armature is pulled toward the plunger. As a result switch 1 attain the state and switch 2 attain the NO state.

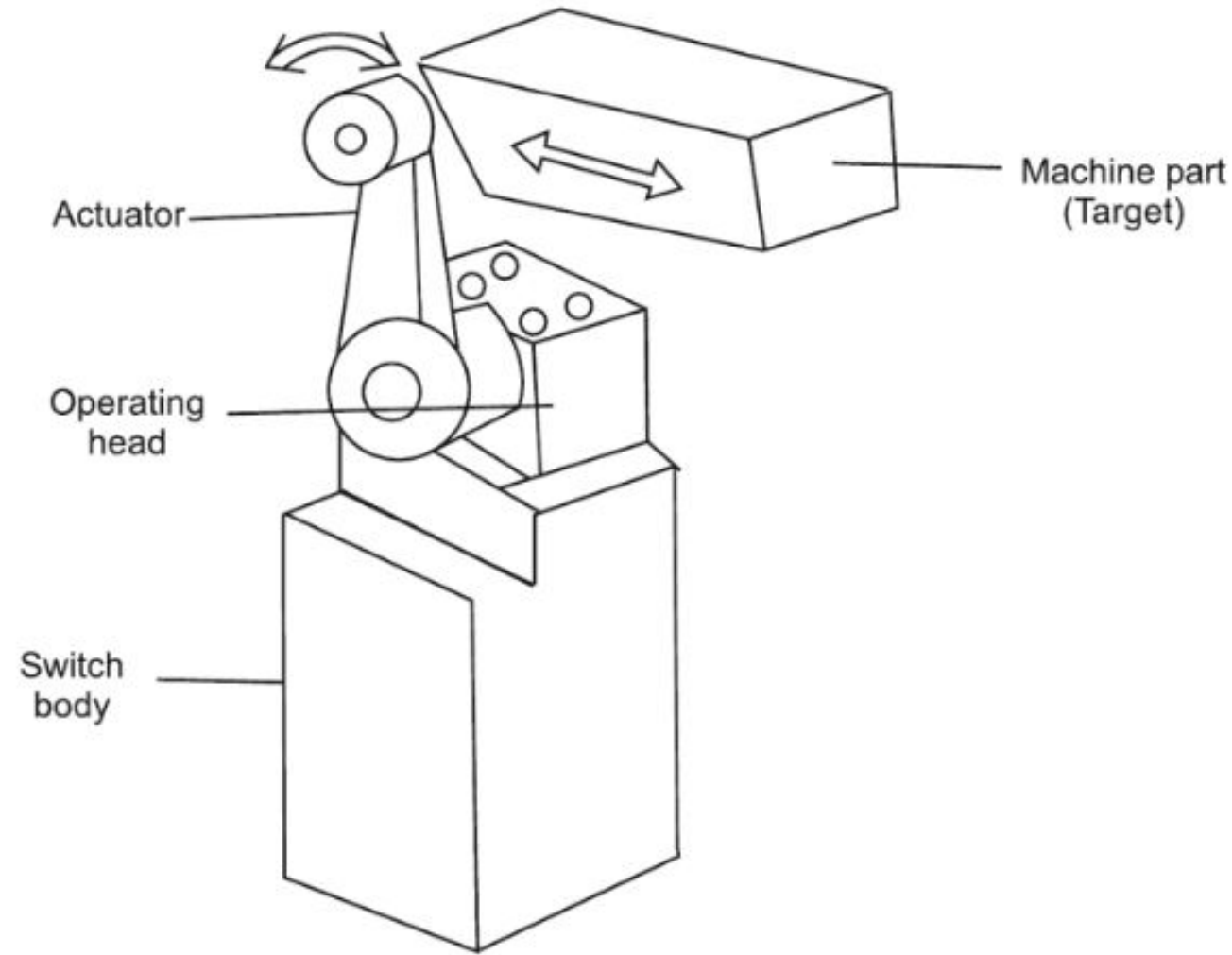
Applications

- Operate controllers in a control system.
- Used in conjunction with transistors in temperature control systems.
- Control the operation of pneumatic and hydraulic valves.
- For delayed switching action time-delay relay are Used.

- Because relays are inductances, they can generate a back voltage when the energizing current is switched off or when their input switches from a high to low signal.
- As a result, damage can occur in the connecting circuit. To overcome this problem, a diode is connected across the relay.
- When the back e.m.f. occurs, the diode conducts and shorts it out. Such a diode is termed a free-wheeling or fly back diode.

LIMIT SWITCH

- A limit switch is a **switch operated by the motion of a machine part or the presence of an object.**
- A limit switch can be used for controlling machinery as part of a control system, as a safety interlock, or as a counter enumerating objects passing a point.



LIMIT SWITCH

- Before the external force is applied the actuator will be in a free position. On contact with the target the actuator lever will travel a certain distance without actuation which is known as **pre-travel**.
- In the operating position the limit switch changes from a normally open (NO) state to a normally closed (NC) state.
- The actuator can travel a certain distance in the NC position beyond the operating point, safely, which is known as **over travel**.
- The release position is achieved between the operation state and the normal state during which the actuator makes the differential travel.
- Between the release position and the free position the distance traveled is the **release travel**.
- The limit switches are used in machine tools to make and break contacts as the machine comes in contact with the actuator of the switch.
- The limit switches are available in different shapes and designs which are selected to suit the application.

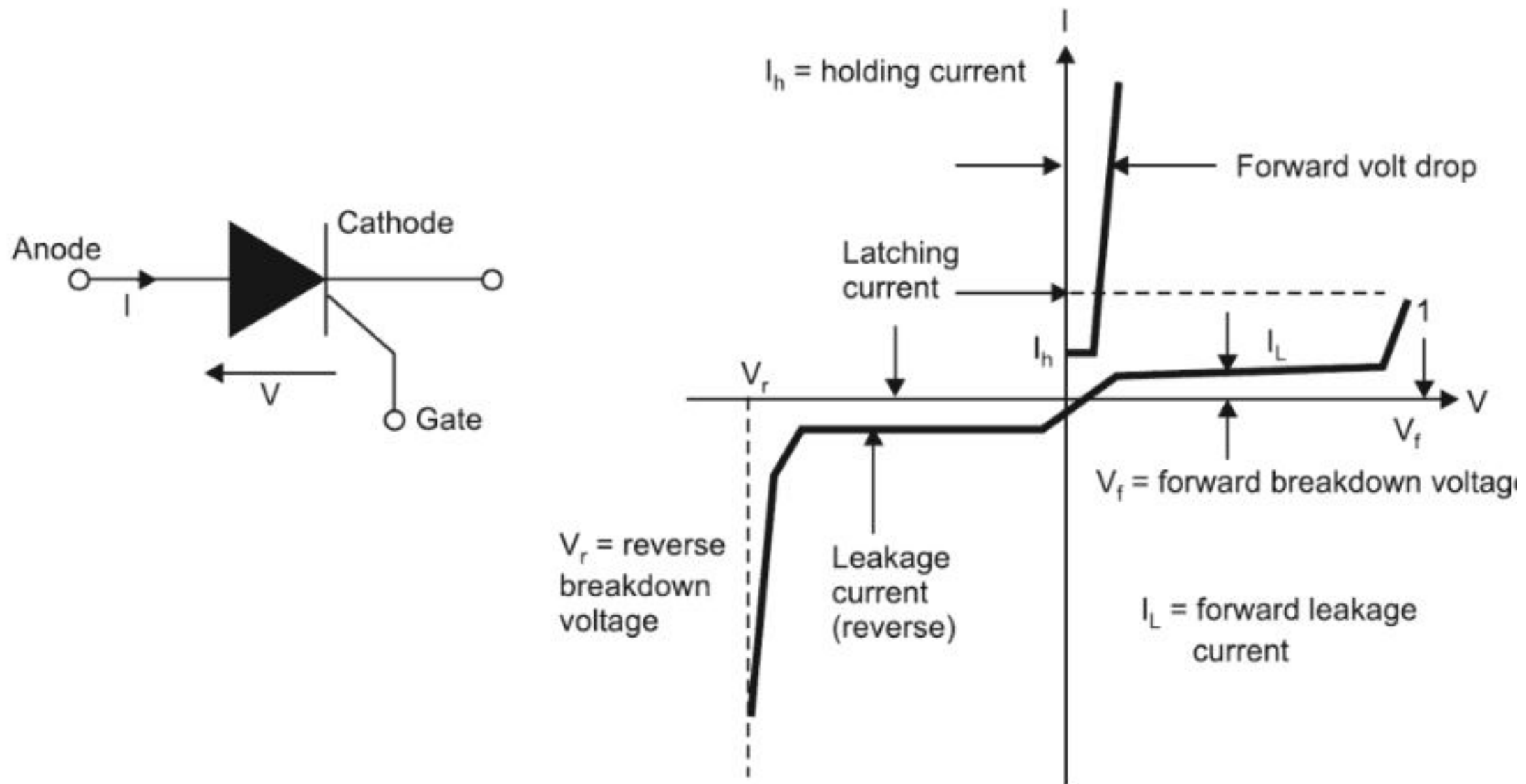
Solid-state switches

To electronically switch the circuits, the following principles of solid-state power switching devices are used:

1. Diodes
2. Thyristors
3. Triacs
4. Transistors (Bipolar)
5. Power MOSFETs

THYISTORS

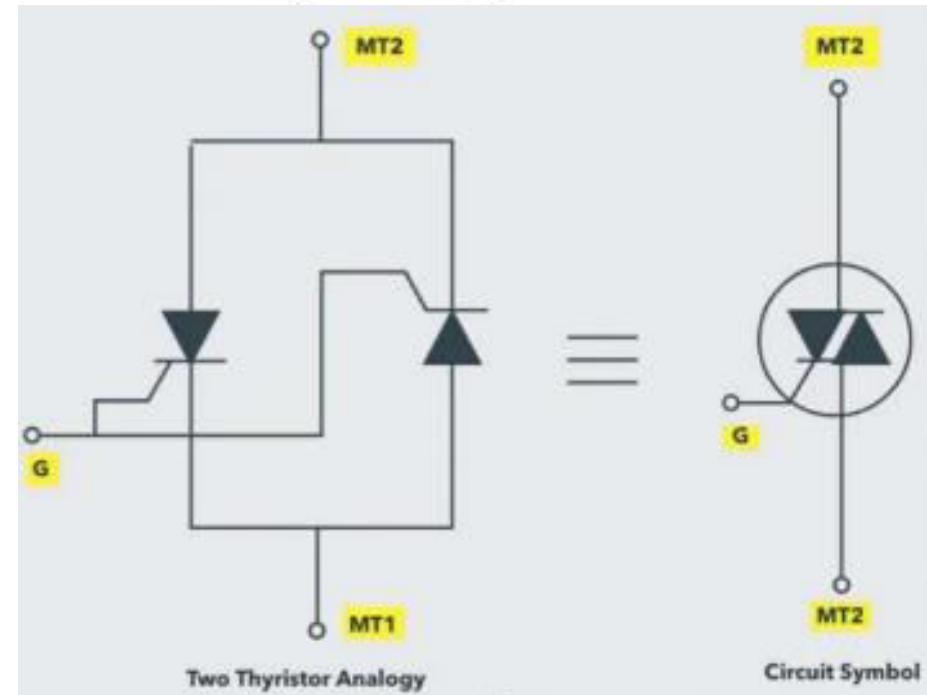
- The other name for a thyristor is Silicon Controlled Rectifier (SCR), and it can be considered **a diode with a gate**.
- With the help of the gate the thyristor can be switched in the reverse bias also.
- When the gate current is zero the thyristor acts as a diode that is unidirectional. Figure 1 shows the characteristics of a thyristor in general.
- Even when forward biased the current is negligible with the forward leakage current.
- At a high forward breakdown current the thyristor switches on, allowing the flow of current, and the voltage drops down to a low value.
- The higher the gate current the lower the breakdown voltage.
- Hence, the switching of the thyristor depends on the gate current. Figure 2 shows the switching characteristics of a thyristor.



Thyristor characteristics.

TRIAC

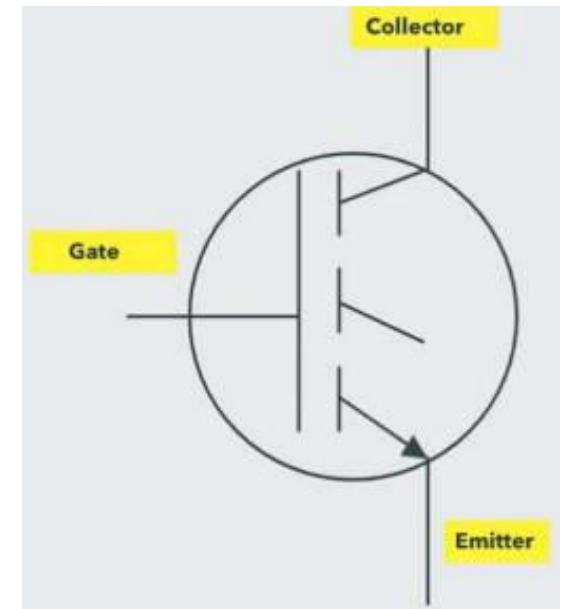
- [Triac](#) (or **TRI**ode **AC**) switch is a bidirectional switching device, which is an equivalent circuit of two back to back SCRs connection with one gate terminal.
- Its capability to control AC power in both positive and negative peaks of the voltage waveform often makes these devices to be used in motor speed controllers, light dimmers, pressure control systems, motor drives and other AC control equipment.



Transistors (Bipolar)

IGBT

- IGBT (Insulated Gate Bipolar Transistor) combines the several advantages of bipolar junction power transistor and power MOSFET. Like a MOSFET, it is a voltage controlled device and has lower ON state voltage drop (less than that of MOSFET and closer to power transistor).
- It is a three terminal semiconductor high speed switching device. These terminals are emitter, collector and gate.



MOSFETs

The MOSFET is the most common type of transistor today. Their primary use is **to control conductivity, or how much electricity can flow, between its source and drain terminals based on the amount of voltage applied to its gate terminal.**

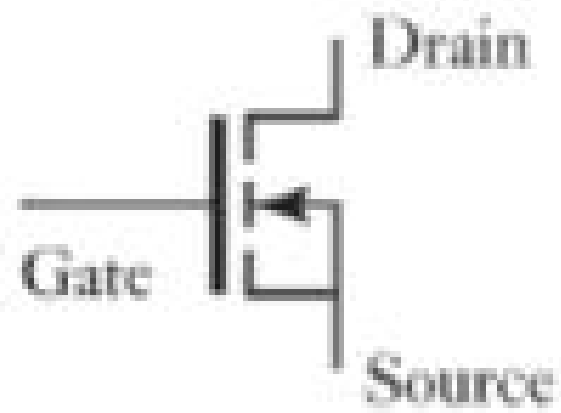
MOSFETs (Metal-Oxide Silicon Field-Effect Transistors) come in two types, the n-channel and the p-channel.

Figure 9.12(a) and (b) shows the symbols.

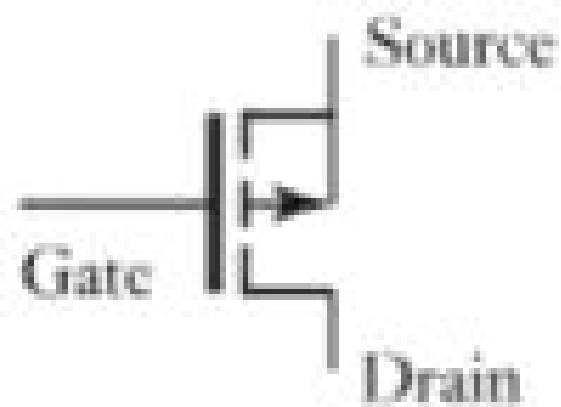
The main difference between the use of a MOSFET for switching and a bipolar transistor is that no current flows into the gate to exercise the control.

The gate voltage is the controlling signal. Thus drive circuitry can be simplified in that there is no need to be concerned about the size of the current.

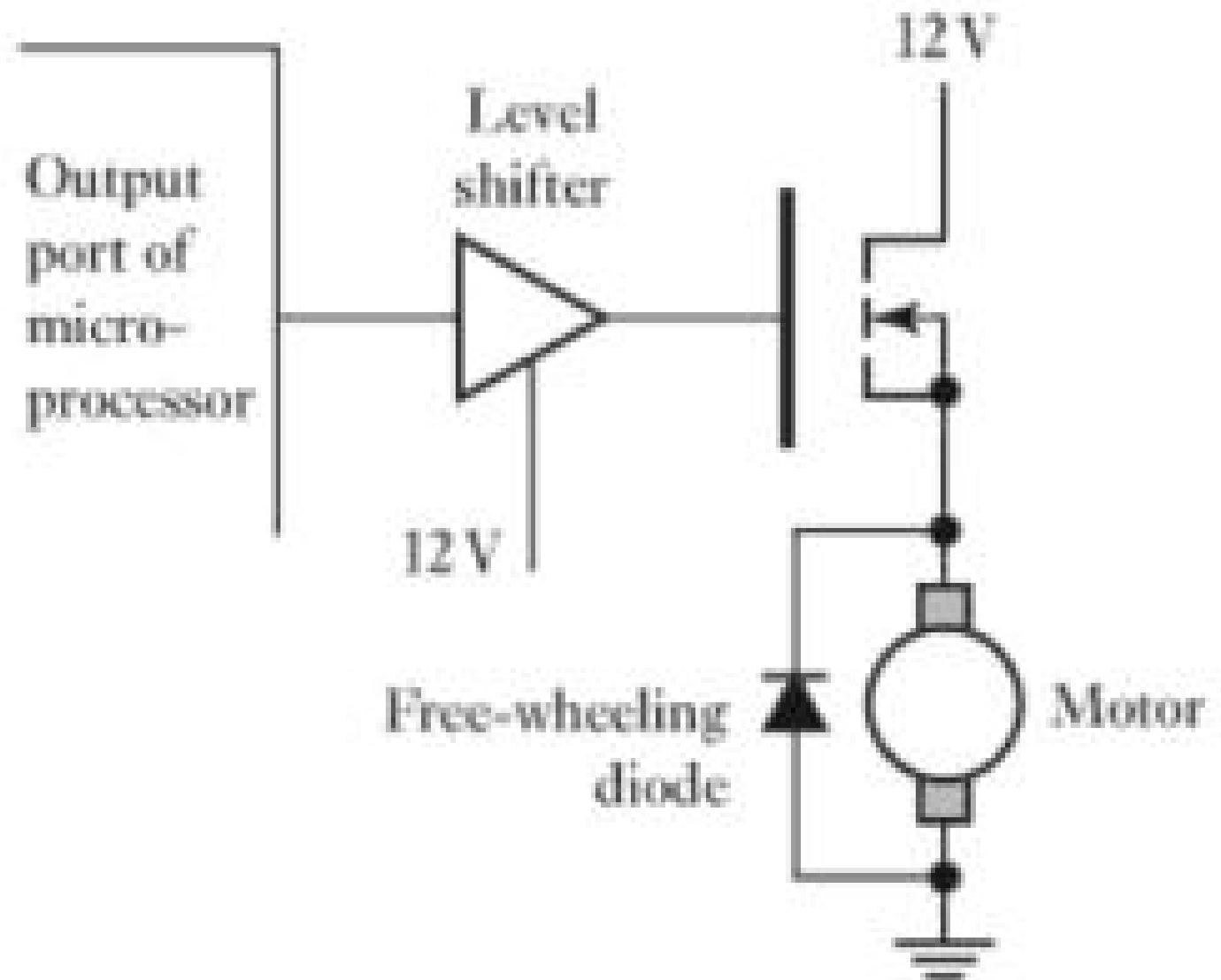
Figure (c) illustrates the use of a MOSFET as an on/off switch for a motor; compare the circuit with that in Figure 9.11 where bipolar transistors are used. A level shifter buffer is indicated, this being to raise the voltage level to that required



(a)



(b)



(c)

MOSFETs: (a) n-channel, (b) p-channel, (c) used to control a d.c. motor.